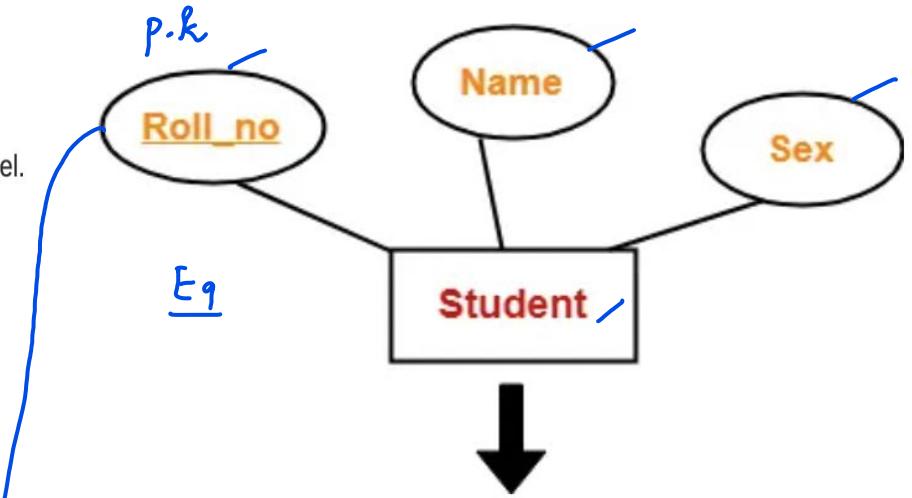


ER DIAGRAM TO RELATIONAL MODEL

Rule-01: For Strong Entity Set With Only Simple Attributes-

A strong entity set with only simple attributes will require only one table in relational model.

- ✓ Attributes of the table will be the attributes of the entity set. ✓
- ✓ The primary key of the table will be the key attribute of the entity set. ✓

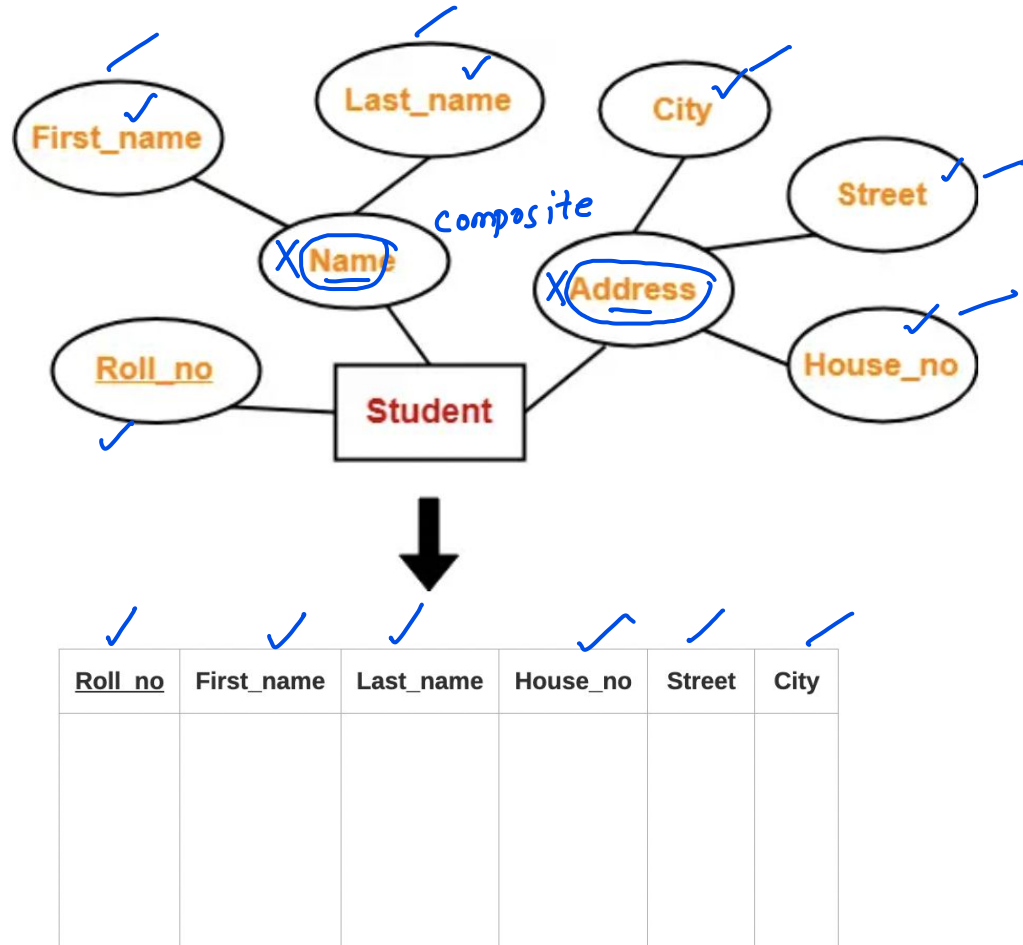


✓ <u>Roll_no</u>	✓ Name	✓ Sex

Schema : Student (Roll_no , Name , Sex)

Rule-02: For Strong Entity Set With Composite Attributes-

- A strong entity set with any number of composite attributes will require only one table in relational model.
- While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.

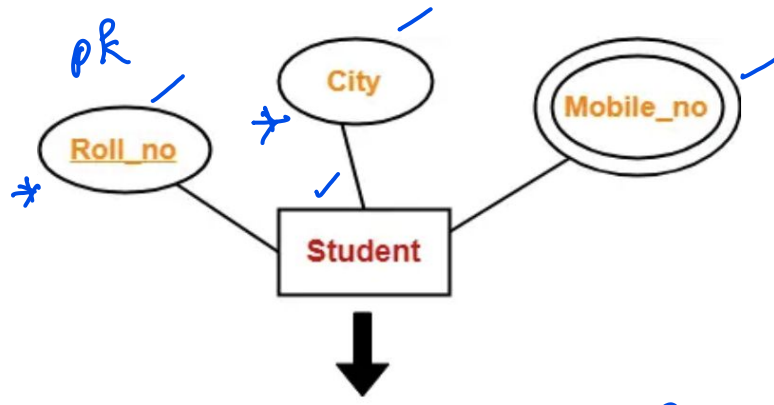


Schema : Student (Roll_no , First_name , Last_name , House_no , Street , City)

✓

Rule-03: For Strong Entity Set With Multi Valued Attributes-

✓ A strong entity set with any number of multi valued attributes will require two tables in relational model.



✓

<u>Roll no</u>	City

Simple attributes /
composite

p.k multivalued

<u>Roll no</u>	Mobile_no

Multivalued

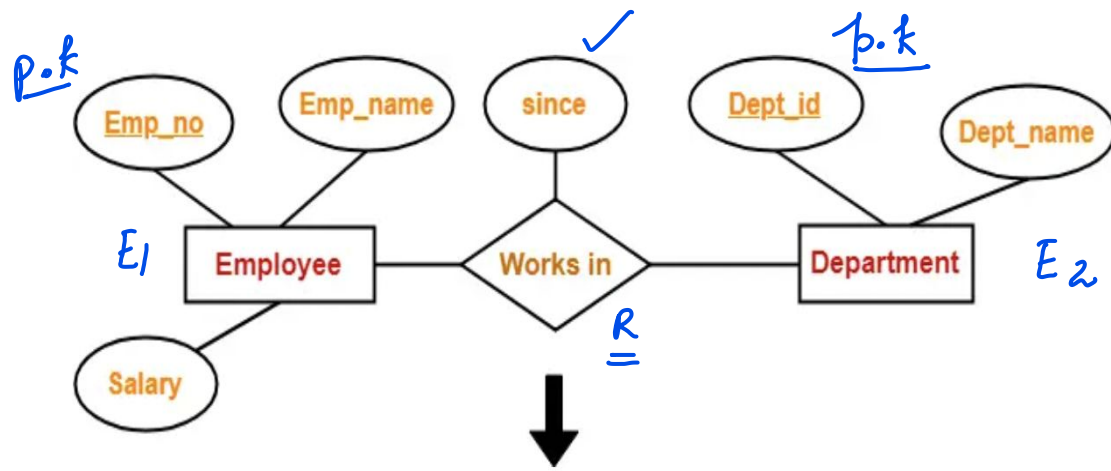
Rule-04: Translating Relationship Set into a Table-

A relationship set will require one table in the relational model.

Attributes of the table are-

- Primary key attributes of the participating entity sets
- Its own descriptive attributes if any.

Set of non-descriptive attributes will be the primary key.



<u>Emp no</u>	<u>Dept id</u>	since

Schema : Works in (Emp no , Dept id , since)

Rule-05: For Binary Relationships With Cardinality Ratios-

The following four cases are possible-

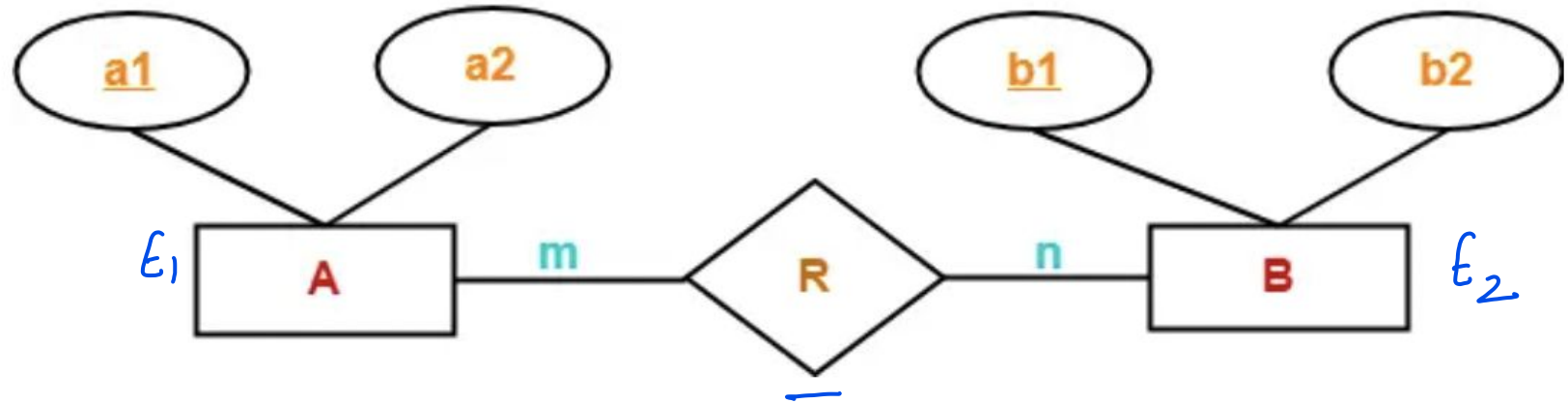
Case-01: Binary relationship with cardinality ratio $m:n$ ✓

Case-02: Binary relationship with cardinality ratio $1:n$ ✓

Case-03: Binary relationship with cardinality ratio $m:1$ ✓

Case-01: For Binary Relationship With Cardinality Ratio m:n

many to many



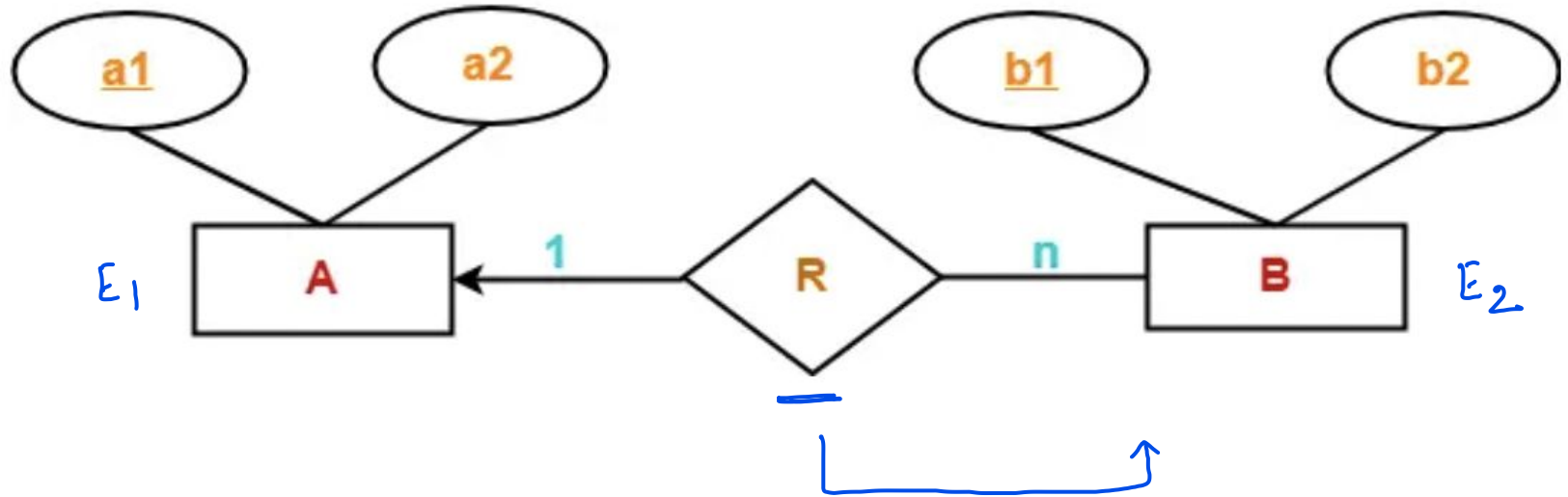
Here, three tables will be required-

1. $A(\underline{a1}, a2)$
2. $R(\underline{a1}, \underline{b1})$
3. $B(\underline{b1}, b2)$

→ R separate table

Case-02: For Binary Relationship With Cardinality Ratio 1:n

1 to many



Here, two tables will be required-

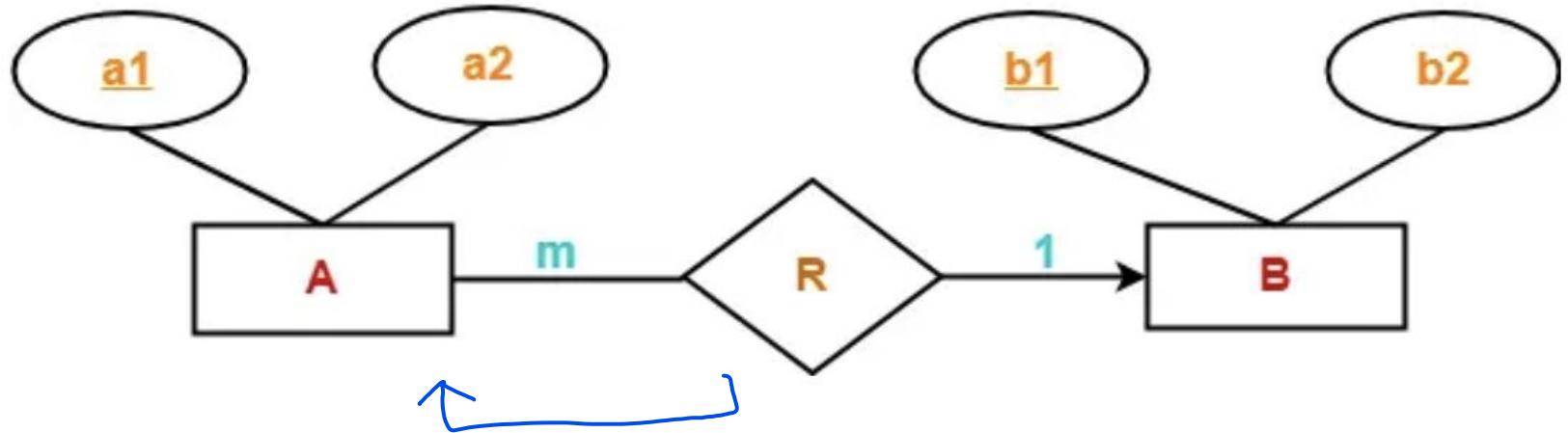
1. $A(\underline{a1}, a2)$

2. $BR(a1, \underline{b1}, b2)$

$\hookrightarrow A$ p.k
 $\checkmark$
 $\checkmark$

Case-03: For Binary Relationship With Cardinality Ratio m:1

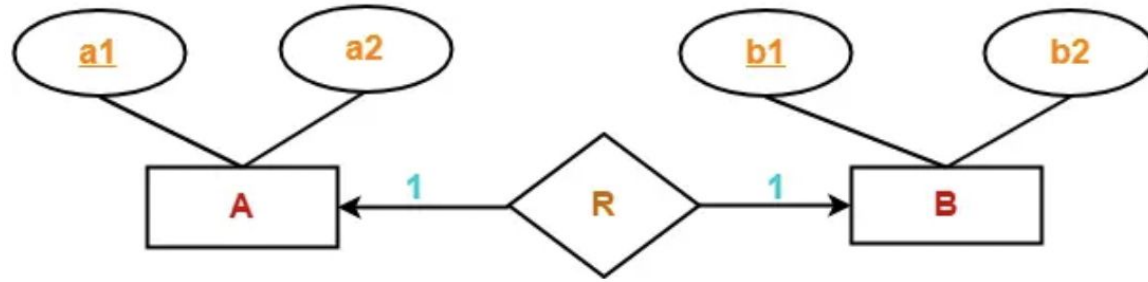
many to one



Here, two tables will be required-

1. $AR(\underline{a1}, a2, b1)$ *B.O.K*
2. $B(\underline{b1}, b2)$

Case-04: For Binary Relationship With Cardinality Ratio 1:1



Here, two tables will be required. Either combine 'R' with 'A' or 'B'

Way-01:

1. AR (a1 , a2 , b1)
2. B (b1 , b2)

Way-02:

1. A (a1 , a2)
2. BR (a1 , b1 , b2)

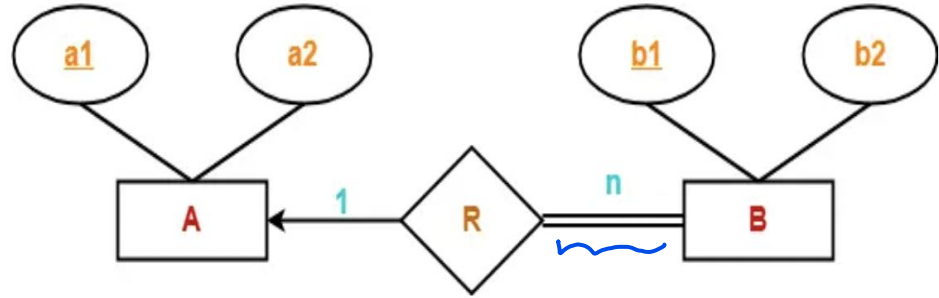
Thumb Rules to Remember

While determining the minimum number of tables required for binary relationships with given cardinality ratios, following thumb rules must be kept in mind-

- For binary relationship with cardinality ratio $m : n$, separate and individual tables will be drawn for each entity set and relationship.
- For binary relationship with cardinality ratio either $m : 1$ or $1 : n$, always remember “many side will consume the relationship” i.e. a combined table will be drawn for many side entity set and relationship set.
- For binary relationship with cardinality ratio $1 : 1$, two tables will be required. You can combine the relationship set with any one of the entity sets.

Case-01: For Binary Relationship With Cardinality Constraint and Total Participation
Constraint From One Side-

1 : many



$A - R - B$

- Cardinality constraints will be implemented as discussed in Rule-05.
- Because of the total participation constraint, foreign key acquires **NOT NULL** constraint i.e. now foreign key can not be null.

Because cardinality ratio = 1 : n , so we will combine the entity set B and relationship set R.

Then, two tables will be required-

1. A (a1 , a2) ✓

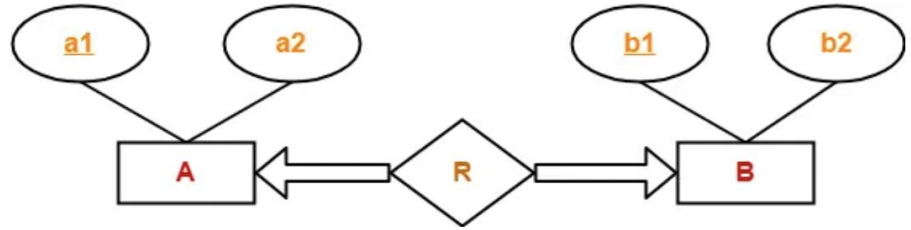
2. BR (a1 , b1 , b2) ✓

$a1 \rightarrow A \text{ pk}$ $\frac{BR}{f.k}$

Because of total participation, foreign key a1 has acquired **NOT NULL** constraint, so it can't be null now.

**Case-02: For Binary Relationship With
Cardinality Constraint and Total
Participation Constraint From Both
Sides-**

If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using only single table.

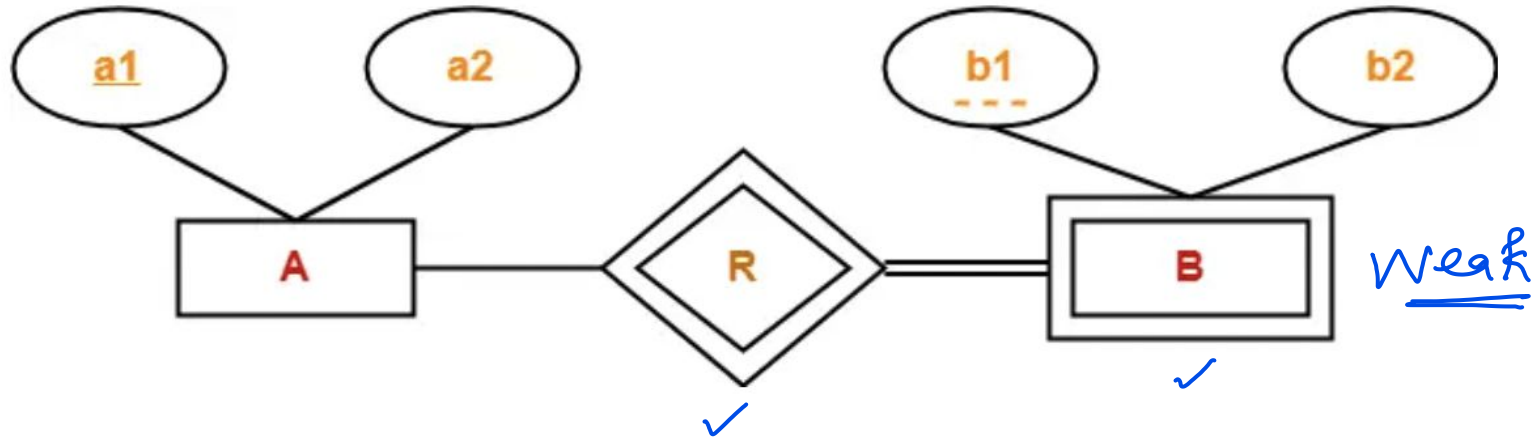


Here, Only one table is required.

- ARB (a1 , a2 , b1 , b2)

Rule-07: For Binary Relationship With Weak Entity Set-

Weak entity set always appears in association with identifying relationship with total participation constraint.



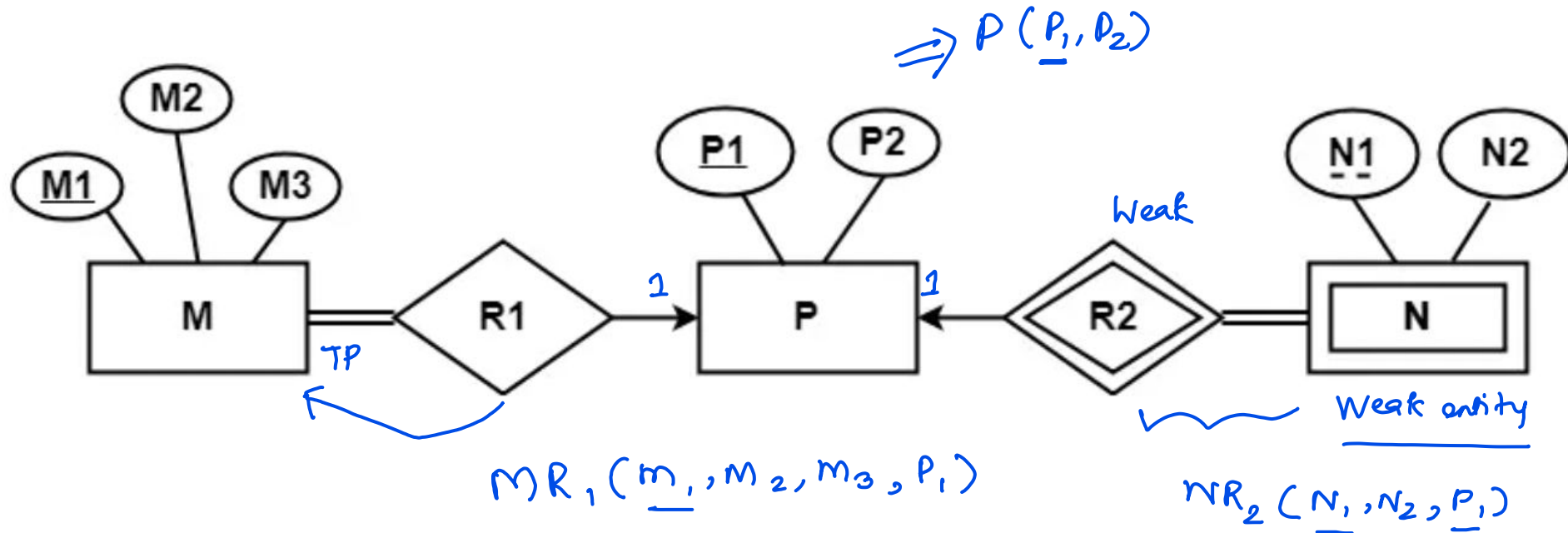
Here, two tables will be required-

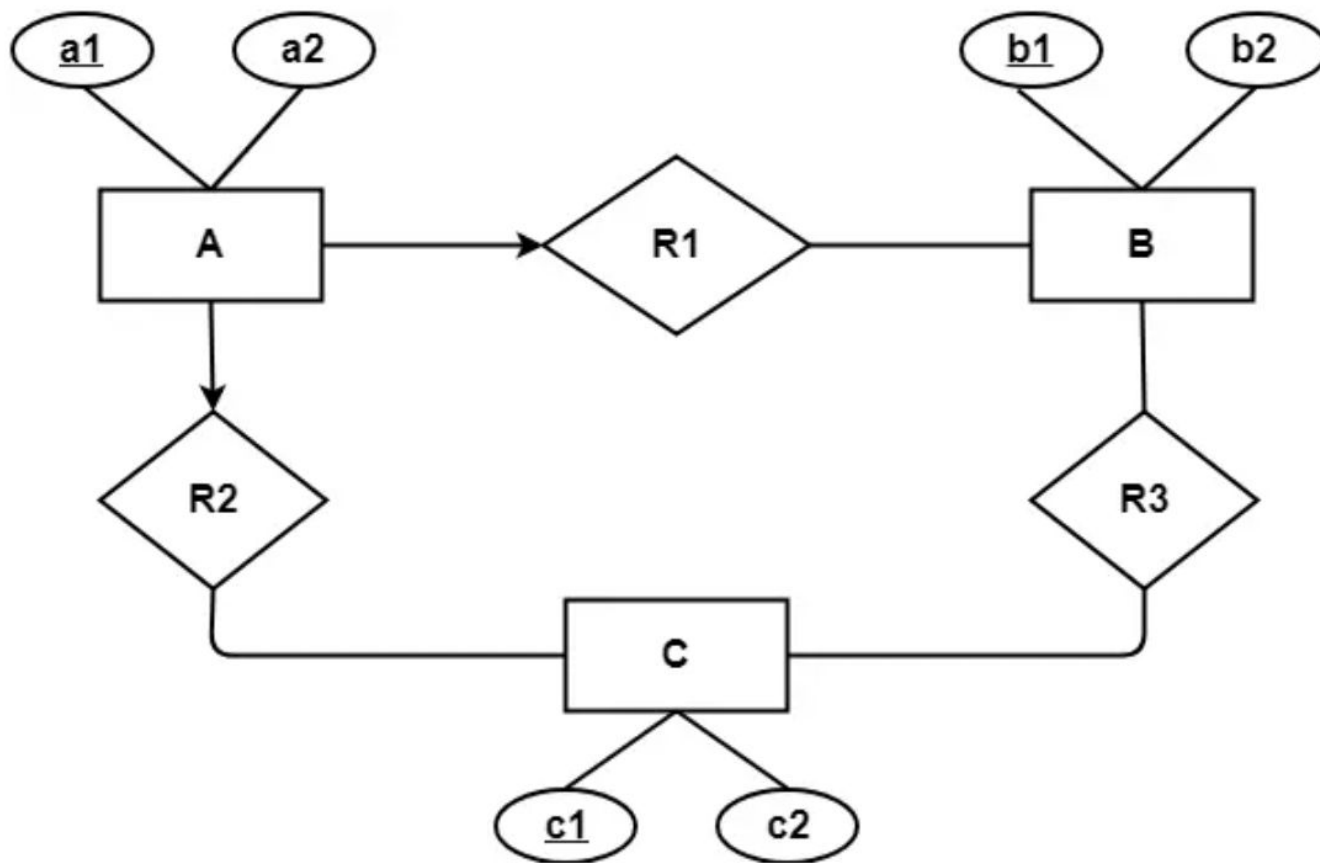
1. A (a1 , a2)

2. BR (a1 , b1 , b2)

Problem-01:

Find the minimum number of tables required for the following ER diagram in relational model-





$AR_1R_2(\underline{a1}, a2, b1, c1)$

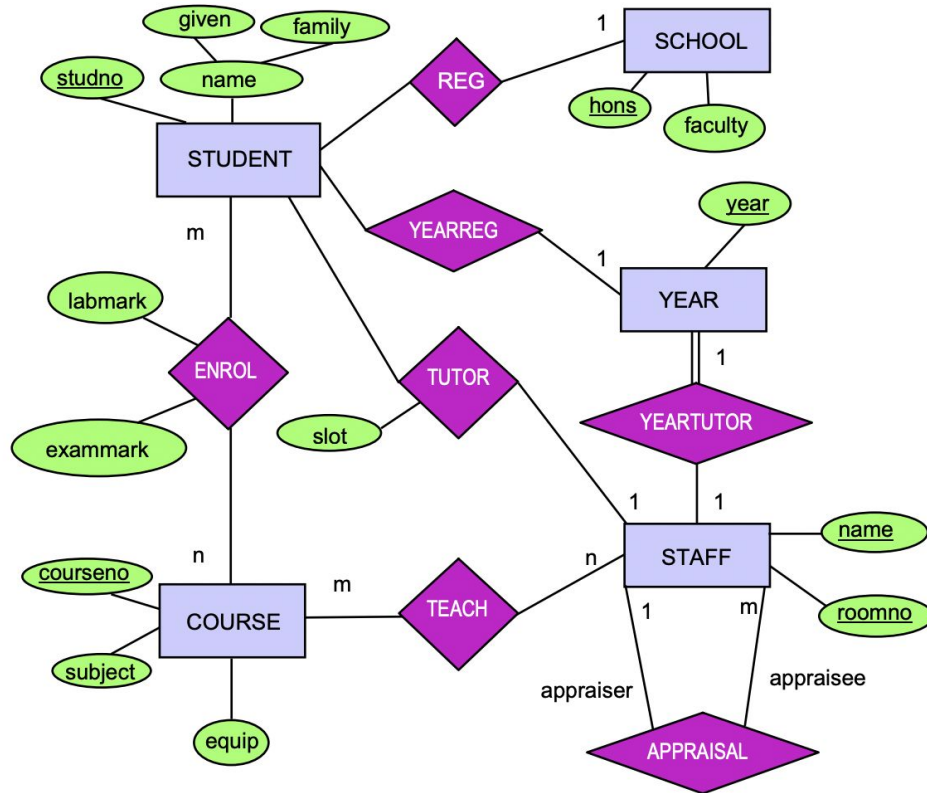
$B(\underline{b1}, b2)$

$C(c1, c2)$

$R_3(\underline{b1}, c1)$

—x—

Translation of the University Diagram



STUDENT

(studno, givenname,
familyname, hons,
tutor, tutorroom, slot, year)

ENROL(studno, courseno,
labmark,exammark)

COURSE(courseno, subject, equip)

STAFF(lecturer,roomno,
appraiser, approom)

TEACH(courseno, lecturer,lecroom)

YEAR(year, yeartutor, yeartutorroom)

SCHOOL(hons, faculty)